

Vibrations and Waves in Physics

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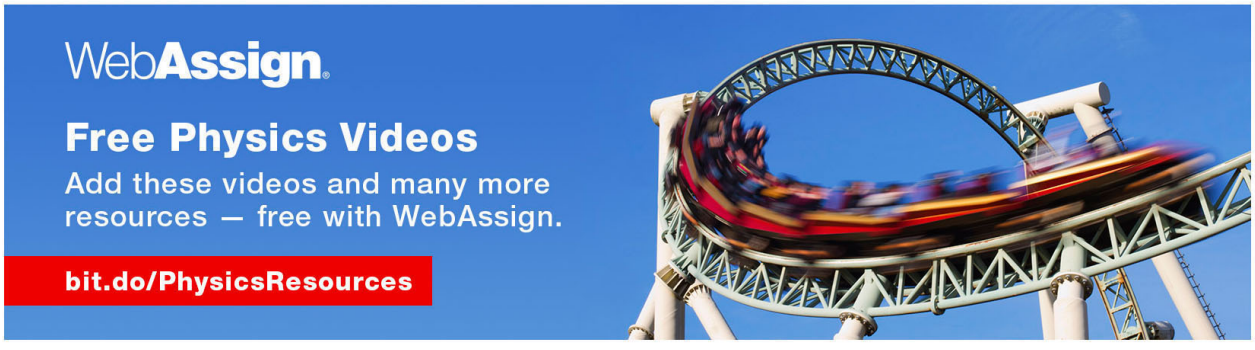
The Physics of Vibrations and Waves, 5th ed.
Am. J. Phys. **69**, 922 (2001); 10.1119/1.4765685

Vibrations and Waves in Physics, 3rd ed.
Am. J. Phys. **69**, 922 (2001); 10.1119/1.4765683

The Physics of Vibrations and Waves, Fourth Edition
J. Acoust. Soc. Am. **94**, 3529 (1993); 10.1121/1.407174

Physical Interpretation of the WKB or Eikonal Approximation for Waves and Vibrations in Inhomogeneous Beams and Plates
J. Acoust. Soc. Am. **48**, 275 (1970); 10.1121/1.1912125

An Introduction to the Physics of Vibrations and Waves
Am. J. Phys. **32**, 66 (1964); 10.1119/1.1970101



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Landau levels, paramagnetism and diamagnetism, the Zeeman effect, hybrid orbitals, van der Waals forces, isotope shift, and energy bands of electrons in solids. Among other outstanding complements are discussions of the chemical bond (H_2^+) and the virial theorem, a treatment of the effect of sinusoidal perturbations: both the linear and nonlinear response of a two-level system and the oscillations of a system between two discrete levels when the perturbation is resonant. Finally, we have discussions of many-electron atoms, a thorough treatment of the levels of helium (configurations, terms, multiplets), and the theory of the properties of an electron gas, with applications to solids.

The book is very well translated. The typography is excellent and errata are notably absent. Illustrations are clear and imaginatively conceived; outstanding examples are the representation of wave packets in Fig. 4 of Chap. I and the drawing of interference effects in a wave packet hitting a potential step (Fig. 1 of Complement J_I).

As an almost encyclopedic resource work, the Cohen-Tannoudji/Diu/Laloë volumes will be extremely valuable for instructors of quantum-mechanics courses and for graduate students. Tailoring a course to this text may be more difficult and would require a revised approach that might prove refreshing. The work is sophisticated: Dirac notation is introduced already in the second chapter, and used throughout; the linear harmonic oscillator is worked out in terms of creation and destruction operators, and Hermite functions are relegated to a complement. But it should be possible to construct a somewhat advanced first course around the work—say, a senior course for strong students who have already gone through a book such as Eisberg's *Quantum Physics*—and the rewards would be high.

Bernd Crasemann received his Ph.D. in nuclear physics from the University of California, Berkeley, in 1953. At present Professor of Physics and Chairman of the Department in the University of Oregon; his research interests are in atomic inner-shell processes and the application of synchrotron radiation to experiments in atomic physics.

Applied Acoustics. G. Porges. 180 pp. Halsted, New York, 1977. Price $16.95. (Reviewed by Thomas D. Rossing.)

One of the most notable gaps in the textbook marketplace is a modern textbook on acoustics at the intermediate or advanced undergraduate level, despite the fact that many such courses at this level are now being offered in physics, mechanical engineering, and electrical engineering departments. A quick glance at this book gives one the impression that it might be an appropriate textbook for an introductory acoustics course at the intermediate level, but further reading raises some doubts.

The book first develops the wave equations for plane waves, spherical waves, and cylindrical waves, and defines basic acoustical terms, including power, pressure, and intensity levels. After a brief chapter on loudness, the transmission of sound through walls, through ducts, and through the atmosphere is discussed. Room acoustics, radiation of sound, noise sources, and vibration isolation are other subjects treated.

Although the choice of topics is reasonable, the coverage of them is uneven. Several important topics (e.g., masking, timbre, and reverberation) are given only a brief descriptive paragraph or two. The chapters on room acoustics, radiation of sound, and noise sources are especially disappointing in this respect.

The number of incorrect or questionable statements appears to be slightly greater than average. For example, one is told (p. 11) that octave band 1 "extended slightly" goes from 20 to 75 Hz and has a midfrequency of 53 Hz; on the same page, it states that middle C of a piano is tuned to 256 Hz (which would put A at 431 Hz). "Standard temperature and pressure" is given as 20 °C (p. 18). The impression is given that when two tones are sounded together the sum-frequency tone will have the same loudness as the difference-frequency tone (p. 48), and it is stated that "for two sounds of the same frequency, the louder one has the lower pitch" (p. 47).

In spite of its shortcomings as a textbook, however, this book does contain a lot of useful information, and it is a desirable book to have in the library. It is reasonably enough priced to recommend it for most university libraries. Meanwhile we wait for publication of an acoustics text destined to set the standards for intermediate-level courses.

Thomas Rossing is Professor of Physics at Northern Illinois University. He has taught courses in acoustics at both the graduate and undergraduate levels and has published several papers on the acoustics of musical instruments.

Vibrations and Waves in Physics. Iain G. Main. 336 pp. Cambridge University, 1978. Price: $37.50 (cloth); $8.95 (paper). (Reviewed by Peter J. Collings.)

This new book successfully provides a strong introduction to wave phenomena and vibratory motion at the undergraduate level. Not only are vibrations and waves properly analyzed mathematically, but examples from all realms of physical phenomena are also used to illustrate the basic results.

The author assumes a working knowledge of trigonometry and calculus, and only some familiarity with differential equation techniques and complex number algebra. Staying well within these guidelines, Main presents a clear and meticulous study of vibrations (including anharmonic and coupled vibrations) and waves (covering dispersion, boundary effects, and diffraction in addition to simple nondispersive traveling waves). After each important result come excellent examples from mechanics, electricity and magnetism, fluids, and thermodynamics. These examples are not presented as cases which happen to illustrate a general result; they constitute an in-depth look at certain physical situations. At least half the book deals with these excerpts from various subfields of physics. Thus a valuable introduction to the subject is achieved, together with an exceptional chance to observe the importance of vibrations and waves throughout the discipline of physics.

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The only question in the mind of this reviewer is where in the undergraduate curriculum Main's text would be most useful. Certainly it could be used on the sophomore level; yet for students to appreciate fully the examples, they will need more than an introductory knowledge of electromagnetism and thermodynamics. Therefore seniors could also gain from a study of *Vibrations and Waves in Physics,* for it would broaden and help to integrate their knowledge of undergraduate physics. In either role, the book represents a masterful accomplishment.

Waves: A Mathematical Approach to the Common Types of Wave Motion (2nd ed.). C. A Coulson and A. Jeffrey. 229 pp. Longman Group, London, 1977. Price: $9.50 (paper).

This book is a revised and expanded edition of Coulson's classic 1941 book on waves. Although some new topics in the theory of linear waves are included, the major change is the addition of a chapter on nonlinear wave theory. Much of the original structure and writing is preserved.

Although the book claims to be an elementary consideration of wave phenomena, this statement can be misleading. For example, a knowledge of multivariable calculus and specialized functions is assumed from the beginning. Results from fluid dynamics and electromagnetism are also used at various points in the text. It is clear that the authors are directing this book at a fairly sophisticated audience.

It is not difficult to understand why this book is a classic. The study is mathematical from the start, containing arguments concerning necessary boundary conditions, proofs of uniqueness, and alternative solutions. The emphasis is on the mathematical solution to problems involving waves, and not on the physical significance of the results. For this reason, the important applications sometimes come at a fast pace, with little concern for the development of physical insight through numerical calculations. Nevertheless, this book will always be extremely valuable to any physicist who desires a rigorous solution to a problem in waves.

Peter J. Collings received his Ph.D. from Yale. He is a member of the Physics Department at Kenyon College.

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